MATH X314 – Data Visualisation

Assessment 4B: Project Portfolio

Has Australian's Three-Decade safety Success Ended? The reversal story – who, where and why?

Submitted by

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Introduction

Australian road users used to be safer until 2020 as government followed strict road safety rules – seatbelt laws, random breath tests, drink driving tests, speed cams, mobile cams, safer cars and road improvements made Australia successful in preventing road fatalities. In 1989, the total road deaths were around 2800 and had fallen to 1091 by 2020, more than 50% decline in the last three decades.

Then after Australian roads safety trend turned around as deaths started rising every year since 2020.

The success story got reversed – who, where and why? – is the story.

SECTION-1: The Australian National Road Fatalities Trend

How has the road fatality trend has changed from 1989 to 2025, Does all road users at risk?

1.1 Has Australia's three-decade road safety success ended?

Australian road fatalities fell gradually from 1989 to 2020. Then the year 2020 arrived. The successful road safety trend suddenly got reversed as road fatalities has started climbing up every year since 2020 – nearly 20% rise in five years.



Figure-1: The national trend of road fatalities from 1989-2025

From reversal trend, the questions are why Road fatalities have suddenly started climbing since 2020 as the Government is still implementing the same reforms that made successful in cutting deaths until 2020?

1.2 who are affecting more in reversal rise?

Splitting the road fatalities using road user type provides the better story - whether the decline is same among all groups or any group is affecting harder.

Drivers and passengers' fatalities count remains flat from last 15 years, since they are having some modern safety features in vehicles.

In contrast, the vulnerable road users on foot such as pedestrians and motorcycle user's fatalities started climbing up as they don't have safety doors like vehicle road users.

Safety doors are not the main gap here, vehicles have gone through several safety advancements in terms of air bags, ABS and lane assistance whereas the vulnerable users remain same, creates the gap between three-decade success and reversal.

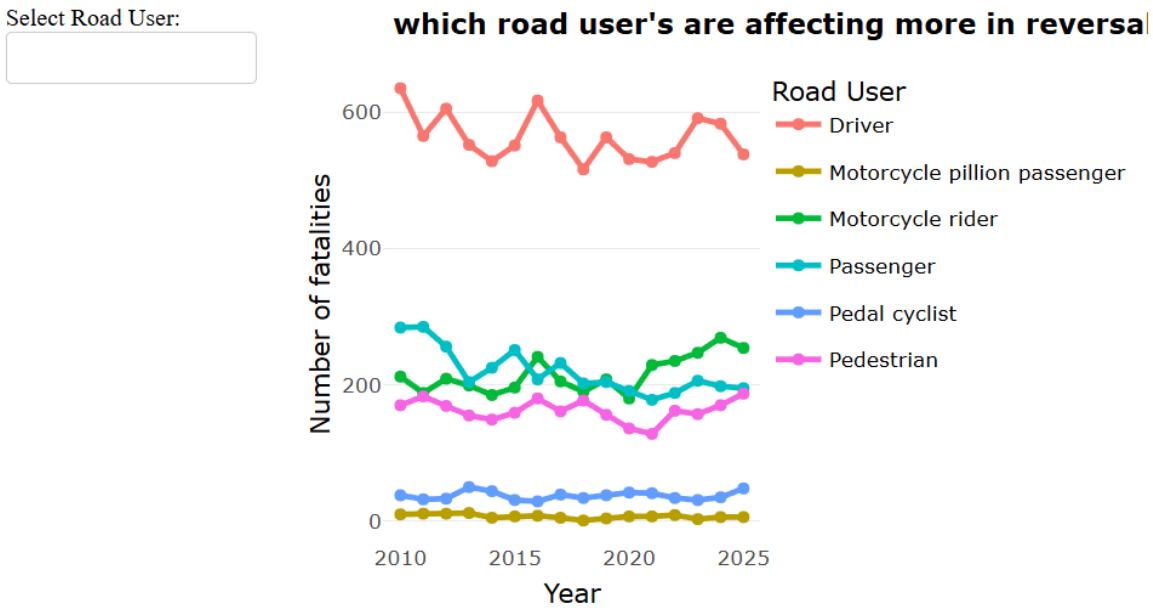


Figure-2: The Australian Road fatalities trend by road user's

Even though, Drivers and passengers are having safety door, they affected more than any other groups as most people use vehicles for travelling. The majority share is drivers and

passengers but in recent rise from above majorly affected groups are vulnerable users. Next question is how deeply these vulnerable groups affected?

1.3 How much percentage does vulnerable occupies in recent rise?

The road fatalities of motor cyclists, pedestrians and pedal cyclists increased around 40%,38% and 14% respectively. Whereas passenger and driver fatalities combinedly increased 3% respectively.

The 94% of road fatalities in reversal trend increment came from the vulnerable users.

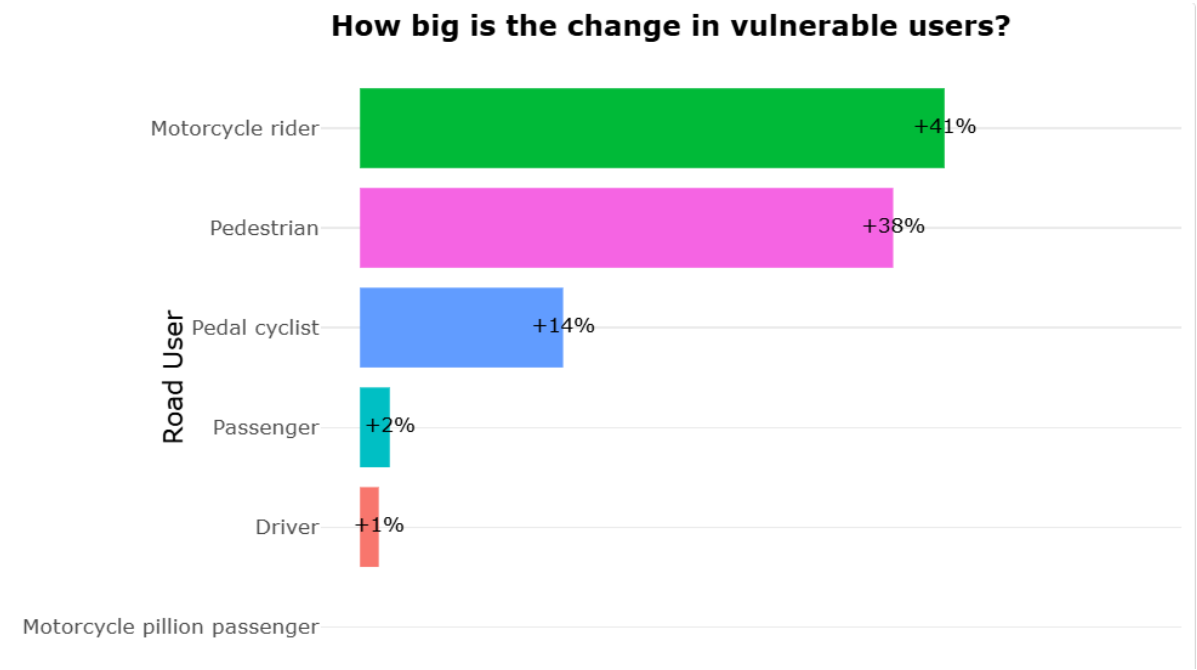


Figure-3: Percentage change in road users' death in reversal period

Almost, the reversal trend is completely dominated by vulnerable users – the interesting question is when and where – geography, location, time and seasons.

SECTION-2: When And Where Fatalities are Higher?

Time Category, Location, Seasons and hidden patterns

2.1 How do states split the road fatalities in reversal trend?

Populations are not same across all states, so for fair comparison- comparing fatalities per 1,00,000 people. The Northern Territory records highest around 14.3 deaths per 100,000 people that is seven times larger than ACT (1.9). Tasmania, Queensland, Western Australia

and South Australia records 6.7,5.2,5.8 and 4.9 respectively and all these states are above national average (4.3).

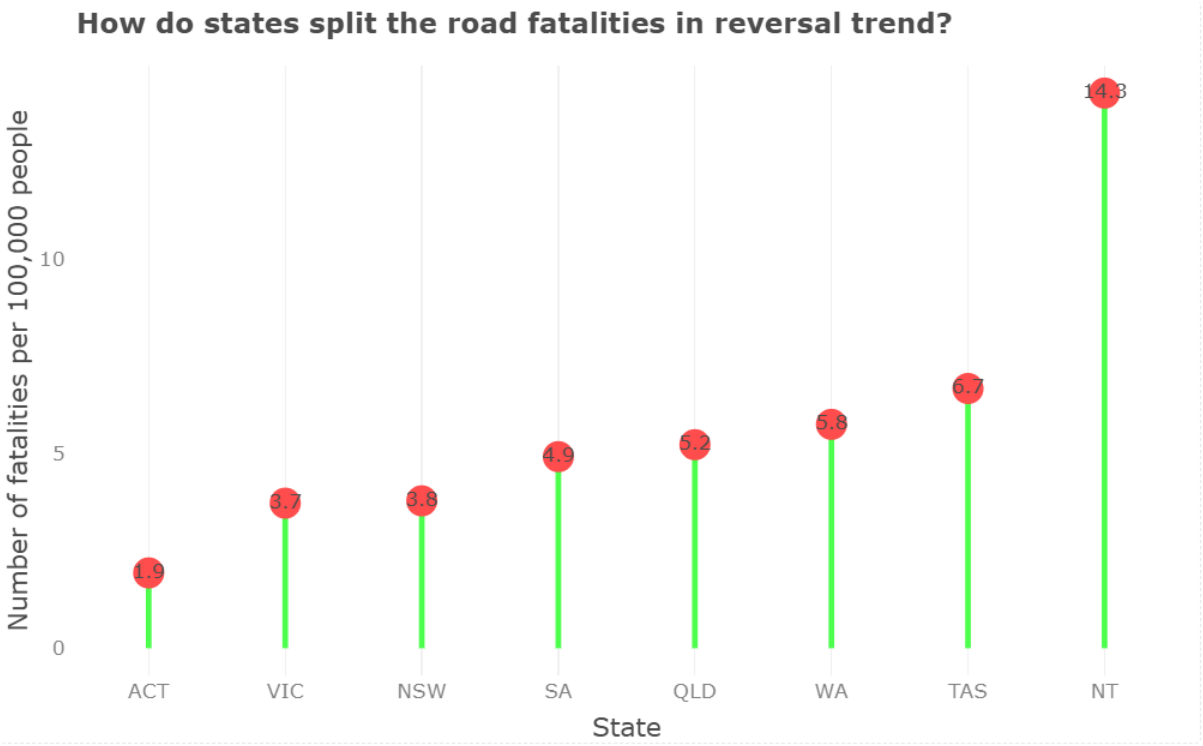


Figure-4: Temporal patterns in reversal trend?

NSW is dangerous state from raw numbers, but it is safest among all states. NT, TAS, WA and QLD have common factors such as more remote areas, higher speed limit zones, long roads and travel long distance for medical services. These factors may be the reason for high road fatalities.

2.2 When Road accidents are more likely to happen – Weekday, Time and seasons

Friday, Saturday and Sunday nights are darkest across all seasons indicates weekend nights are dangerous to travel. Many reasons – low visibility, drink and drive, longer evenings, free time makes people to travel at high speeds to match weekend vibes – makes higher fatalities in these combinations than any other. The remaining combinations remains the same across four reasons as low to medium risk zones.

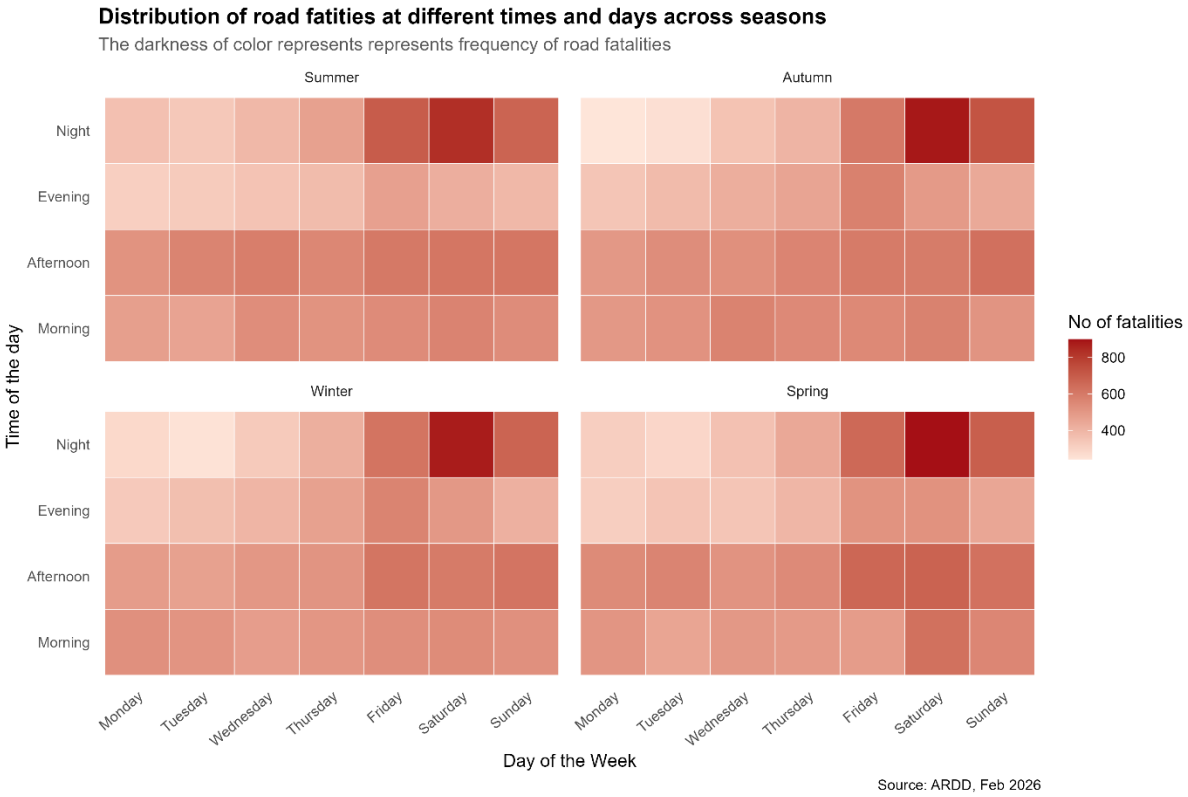


Figure-5: Road fatalities by day and time across seasons

In summer and spring along with weekends the regular times are also darker than other two seasons. Now, we know where deaths are concentrated, to go a little deeper into the story – on what kind of roads these fatalities are higher?

2.3 **which speed limits are dangerous for whom?**

Yes, different road users affected more at different speed zones. Pedestrians and Pedal cyclists at lower speed zones – rural and semi urban areas, whereas drivers and passengers nearly take half of the share in Highways or 110 zones. Overall, the major share across all zones occupied by drivers and passengers as mostly people travel in vehicles.

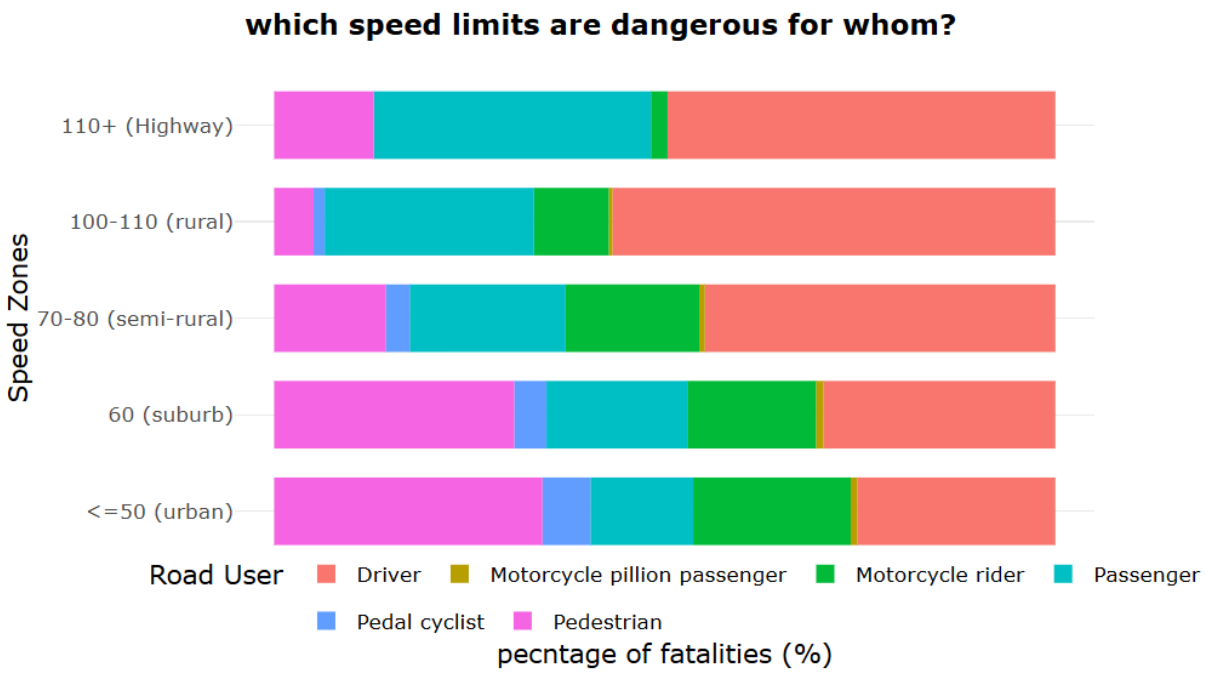


Figure-6: proportion of road users' deaths by speed zones.

This pattern directly tells us that Pedestrians and motorcyclists' deaths are high at urban (<=50) areas. It may be because of over speeding, crossings, different paths and junctions. It is completely road infrastructure problem.

The recent rise we have seen because of 40% increment in vulnerable users' deaths and most of them affected in less than 60 speed zone. Here, we got the direct solution where we need to concentrate more to reduce the recent rise and make the Australian roads safer for vulnerable users.

To complete our story, we need to little deeper and compare the crash types and demographics to take necessary steps to cutdown the road fatalities.

Section-3: which age and gender groups affected more?

single vs multiple and male vs female

3.1 Which crashes are dominant?

The data shows a significant change in the road crash patterns over time. Single-vehicle crashes are common at night whereas multiple crashes at mornings and afternoons.

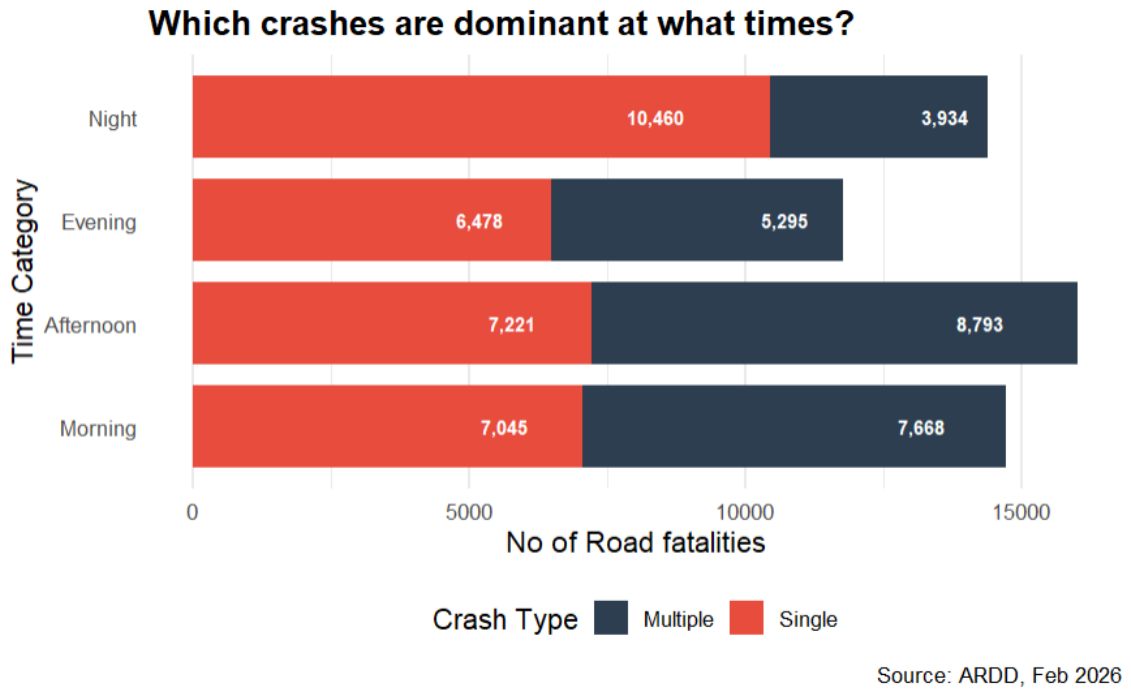


Figure-7: Shows single vs multiple crashes

Single and multiple crashes share equally at evenings. Nights are dangerous for single crashes as they are many factors such as low visibility, speeding's, chance of hitting stationary object is high, empty roads and less traffic. These single crashes completely depend on driver and vehicle condition.

3.2 Which age groups and gender affected more?

Male victims are mostly concentrated at younger to middle - aged - risk taking, racings, over speeding and night driving, whereas females spread across young, middle and old aged. The median and mean are higher for female than male, The black diamond represents mean, in both boxes it sits above median indicate the distribution is right skewed it means a few older victims increases the average.

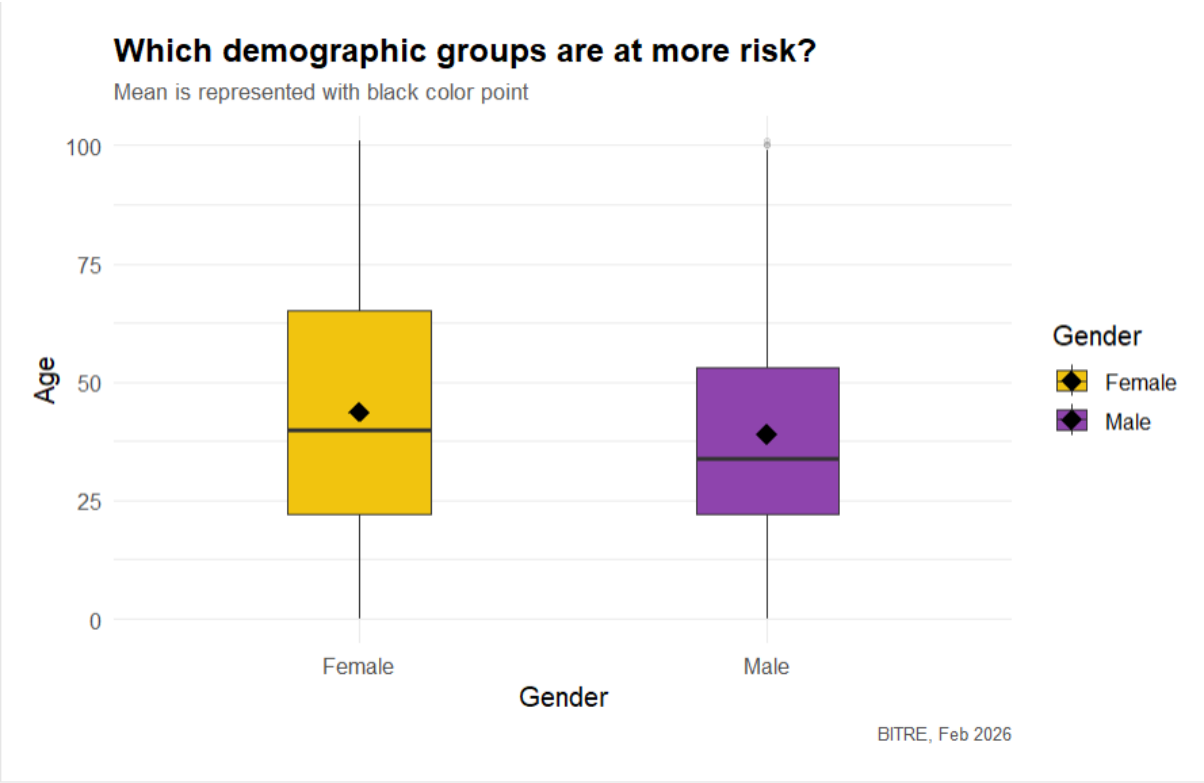


Figure-8: Road fatalities by demographics

Conclusion:

The Australian three-decade road safety success of declining road fatalities ended in 2020. Since, then it has started reversing back by climbing up road fatalities every year. The reversal is mostly affecting the vulnerable road user's – pedestrians and motorcyclists than drivers and passengers. There is no meaningful pattern observed for drivers and passengers but that we have seen in vulnerable users whose deaths nearly 40% increase in last five years. Remote states are nearly carrying 6-10 deaths per 1,00,000 people than other states. By speed limits, Low speed zones are high risk for pedestrians and motor cyclists and high-speed zones for drivers and passengers.

The weekend nights across all seasons especially summer are scary as we have seen more crashes during this combination. Nights dominate with single crashes, mornings and afternoons with multiple crashes. By demographics, young to middle aged male groups are affected harder than any other groups.

These are not just patterns but also tells us the times, geography, location and profiles at are high risk. The road safety strategy for normal road user may miss this highly risk profiles. But a strategy for high risk profile at specific location, time and speed zones save many

Takeaway

If you are an Australian public who drives on Australian roads then this story helps you to know about current situations of Australian road safety – what it have achieved in three decades and what need to achieve in next ten years.

Thirty-years ago, Australian vehicle users were at more risk – road advancements, car safety improvements and government road safety reforms made succeeded in cutting down the road fatalities to half over three decades. The Australia's three-decade success of road safety was built around the vehicle users it was worked but it didn't care about vulnerable users and left behind.

If you are travelling on Australian roads -you may have the safest car contains – airbags, seatbelt, protected metal cage and anti-lock braking system but it may not be same for pedestrians, pedal cyclists and motor cyclists

Now, they are at more risk – the pedestrian who is crossing the road in front of you at 50 km/h, the motor cyclist at intersection.

If Australia has did it once then they can do it again, but you need to be part of this success story just by slowing down the vehicle where pedestrians walk, motor cyclists cross, crossing pedal cyclists.

Slowing down for a second could save a life. That's how the Australia's new success story begins.

Appendix A – Key Decisions in Storytelling.

The ARDD data contains variables such as speed, time, area, crash types, vehicle types, demographics and all the information required for road fatality analysis. Since, ARDD contains enough data, I have different possible options for storytelling. Among them, I chose the reversal story of three-decade success of road safety success. The reason for selecting the reversal story - 2020 is the breakpoint for success story and it is recent enough to make the story and tells a strong story of 30 years success followed by 5 consecutive years of rising road fatalities. The story strong message to Australian public about realistic situation of road safety in Australia and it also tells the story is not same and it is changing from last 5 years. So, instead of choosing whole dataset and I observed different pattern in last 5 years and constructing the graphics which makes sense than making working on whole dataset.

Previously, I had ten graphics covered all the variables from the data set and I used complete data set. It made graphics repetitive and the story is not strong enough. After continuous research I followed the strategy of each graphic should add value to the story than I reduced to my graphs from 10 to 8 covering all the required variables making the story strong. I also divided the story into three sections to make it understandable. I reused few graphics from Assignment 2 and added some new graphics for this reversal story.

I created graphs and narratives from user perspective; non-tech people can clearly understand the situation of Australian road safety from the visual representation which raw data can’t provide. The graphs and narration are simple and understandable.